

5 The kidney is an organ that is involved in osmoregulation and the production of urine.

(a) (i) Which substances are filtered from the blood in the Bowman's (renal) capsule of a healthy individual?

(1)

- ☐ **A** glucose and prothrombin
- ☐ **B** glucose and urea
- ☐ **C** glycogen and urea
- ☐ **D** urea and prothrombin

(ii) Which part of the Bowman's (renal) capsule prevents plasma proteins from being filtered out of the bloodstream?

(1)

- ☐ **A** basement membrane
- ☐ **B** smooth muscle of the capillaries
- ☐ **C** epithelial cells of the collecting duct
- ☐ **D** epithelial cells of the loop of Henlé

(iii) Which transport mechanism is responsible for the uptake of glucose into the cells of the proximal tubule?

(1)

- ☐ **A** endocytosis
- ☐ **B** exocytosis
- ☐ **C** osmosis
- ☐ **D** sodium co-transport

(b) Describe how the loop of Henlé acts as a countercurrent multiplier.

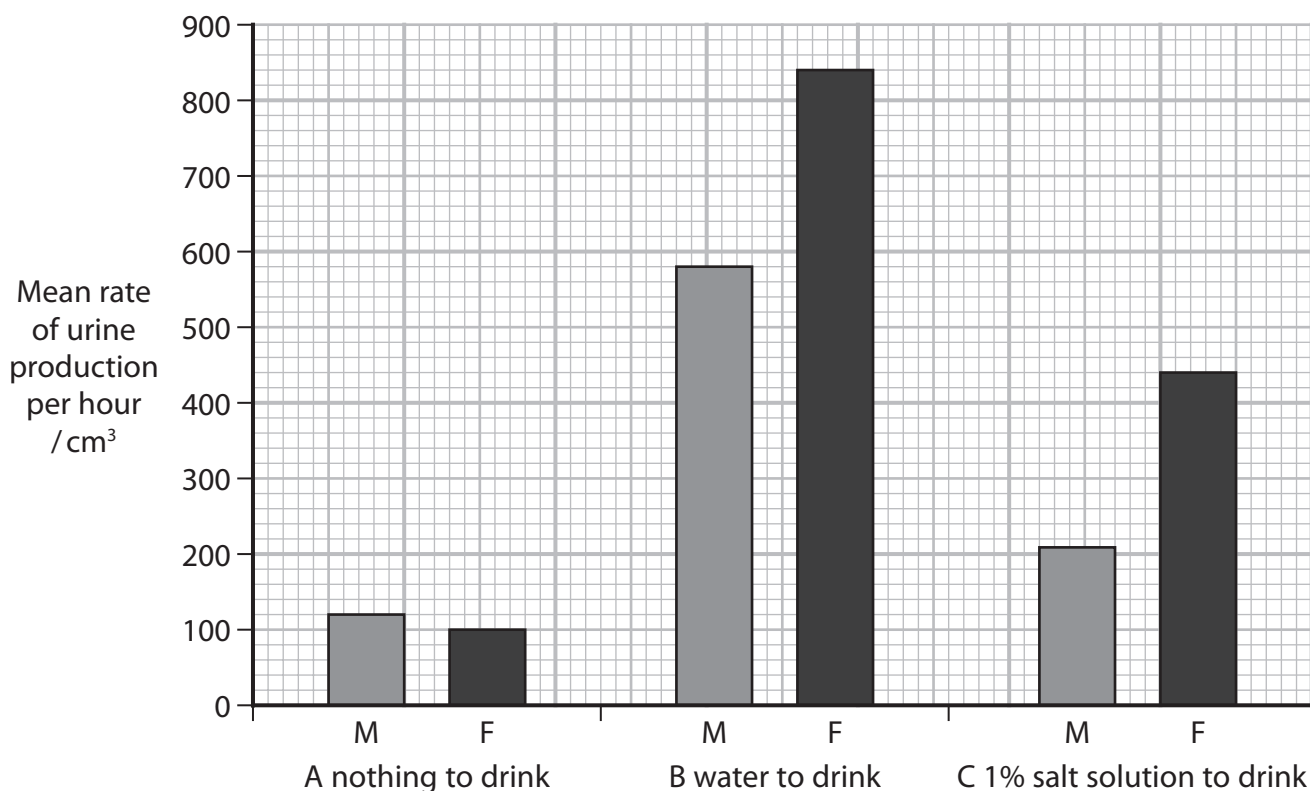
(4)

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- (c) In an investigation, one group of people was given nothing to drink, a second group was given water and the third group was given 1% salt solution to drink.

The mean rate of urine production was recorded for the males and females in each group.

The graph shows the results of this investigation.



- (i) Calculate the ratio of the mean rate of urine production for the females in the three groups.

Express your answer as nothing to drink:water to drink:salt solution to drink.

Give your answer to one significant figure.

(2)

Answer

(ii) Describe **two** conclusions that can be drawn from the results of this investigation.

(2)

(iii) Explain the effect on the mean urine output of the group drinking water.

(2)

(Total for Question 5 = 13 marks)